EMELIO JOSHIA EXAUDI LUMBAN GAOL

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Meteorology graduate (Institute of Technology Bandung) who turns raw environmental and public data into analysis people can act on, using automated Python/SQL pipelines, Power BI and Tableau dashboards, and statistical models. Committed to high personal standards and continuous improvement.

EDUCATION

Institut Teknologi Bandung — Bandung, Indonesia
Bachelor of Science in Meteorology; GPA 3.15/4.00; Semester 7 GPA 4.00/4.00 — Aug 2022 – Sep 2026
- **Thesis:** Quantified land-use impact on peak discharge in DAS Sepaku (IKN): 12 WRF-Hydro runs, 20 years of satellite rain.
- **Award:** National Finalist, IEEEBIG 2026 Business Case Competition, with ECOS, an EV Battery Circular Ownership System.

EXPERIENCE

PT Aneka Tambang Tbk (ANTAM) — Bogor, Indonesia
Environmental Analyst Intern — Jan 2026 – Feb 2026
- Built a Python pipeline merging 20 years of GPM satellite rainfall, IoT rain gauges and extensometers for mine slope safety.
- Found 16 days on which the site's alert SOP missed national early-warning triggers; recommended cumulative-rainfall criteria.
- Documented 10 data-integrity issues; showed a "4x correlation" came from one flagged outlier, preventing a false conclusion.
- Designed a 9-panel review dashboard (HTML/JS, Chart.js) for climatology, extremes, thresholds and data integrity.

Indonesian Agency for Meteorology, Climatology, and Geophysics (BMKG) — Jakarta, Indonesia
Climate Change Analyst Intern — Jul 2025 – Aug 2025
- Downscaled CMIP6 projections from about 100 km to site level (Python, bias correction) for mining climate-risk assessment.

Institut Teknologi Bandung — Bandung, Indonesia
Teaching Assistant, Observational Meteorology & Continuum Mechanics — Sep 2024 – Sep 2025
- Developed practicum modules and led weekly sessions for 60+ undergraduates, including Python data-analysis tutorials.

Laboratory Assistant, Applied Meteorology Laboratory — Apr 2024 – Sep 2026
- Set up an inventory-tracking system for 30+ instruments and safety inductions, cutting gear-retrieval time for researchers.

Research Assistant & Field Surveyor — Jan 2024 – Dec 2024
- Conducted field surveys of residents and measured past flood marks and drainage conditions across the survey area.
- Co-designed and field-tested "Jaring Embun," a low-cost dew-harvesting instrument for arid regions, from prototype to deployment.

PROJECTS

Shopee & Monee Unit Economics | *Python, SEC EDGAR, Excel, Power BI* — Oct 2026
- Parsed 30 quarters of Sea Ltd filings into Excel: Shopee's fee and ad take rate roughly doubled to 11.1% of GMV; EBITDA 0.67%.
- Built a FY2026 EBITDA nowcast and simulator (US$1.04B vs US$1B goal); Monee's 1.0% NPL is 1.6% on year-ago loans.

Procurement Red-Flag Monitor | *Python, web scraping, SQL (DuckDB), scikit-learn, Power BI* — Oct 2026
- Scraped 4,165 tenders from 34 provincial SPSE portals; built 6 SQL red flags, an anomaly model and a Power BI shortlist.
- Found 28% of tenders carry 2+ red flags; single-bidder wins came about 1% below the owner's estimate vs 5–7% with competition.

Indonesia Food Price Monitor | *Python, web scraping, SQL (DuckDB), Power BI, DAX* — Sep 2026
- Automated a weekday pipeline of 864,000 PIHPS prices into SQL and Power BI; chillies drive the price risk (+92% in a year).

LEADERSHIP & ACTIVITIES

Atmosphaira, Meteorology Student Association ITB — Bandung, Indonesia
Head of Research and Development Division — May 2025 – Feb 2026
- Led a 10-person team from hardware prototype and firmware to field deployment of an IoT water-level and rainfall instrument.
- Shipped a real-time web dashboard combining live instrument data with WRF-Hydro river-discharge forecasts.

OSKM ITB 2023, Central New-Student Orientation — Bandung, Indonesia
Mentor — 2023
- Guided new students through their introduction to the ITB campus environment.

SKILLS & CERTIFICATIONS

Languages: Python (pandas, xarray), R (RStudio), SQL, HTML/CSS/JavaScript, Linux/Bash.
BI & Tools: Power BI (DAX), Tableau, MS Excel, Word, PowerPoint, Figma, QGIS, Google Earth Engine, WRF, WRF-Hydro.
Methods: Statistical Modeling, Machine Learning, AI-Assisted Analysis (LLMs), Anomaly Detection, Web Scraping, ETL, Data Visualization, Geospatial Analysis.
Certifications: Coursera: Microsoft Power BI Data Analyst, Google Data Analytics; Forage: Deloitte Data Analytics Job Simulation; WCRP: CARE for SEA & CORDEX-SEA Workshop.
Spoken Languages: Indonesian (native), English (professional working proficiency).
Hobbies: Basketball, badminton, padel, playing music (guitar, piano), chess, gaming (CS:GO, Valorant, FIFA, NBA 2K).